

CSE475 — Task 2: Baseline Models and Proposed Graph Neural Network

NinaPro DB1 EMG + Glove Gesture Classification

Group 04

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Abstract

This report presents Task 2 of a gesture-classification project on NinaPro DB1: a suite of classical machine-learning baselines and a proposed graph neural network (GNN) that represents each of the dataset’s 32 sensor channels (10 EMG + 22 glove) as a node in a correlation-structured graph. All experiments use a subject-independent train/test split (subjects 1–20 train, 21–27 test) to avoid subject leakage. The best baseline, an RBF-kernel SVM, achieves Macro-F1 = 0.7842; the proposed GNN achieves Macro-F1 = 0.3945, underperforming every classical baseline tested. We report full results, describe the proposed architecture and design rationale in detail, and give an honest analysis of why the GNN underperforms, connecting the result back to the label-collision finding from the Task 1 EDA.

1 Problem Setup

1.1 Data and Splitting

Both the baselines and the proposed GNN use the same windowed feature construction established in Task 1: raw 100 Hz EMG (10 channels) and glove (22 channels) signals from all 27 subjects and all 3 DB1 exercises are windowed (200-sample window, 100-sample step, 50% overlap), grouped so no window spans a gesture or repetition boundary, giving 30,936 labeled windows across 23 gesture classes (rest excluded).

The train/test split is **subject-independent**: subjects 1–20 form the training set (22,724 windows) and subjects 21–27 form the held-out test set (8,212 windows). No subject appears in both sets, so the reported performance reflects generalization to entirely new users rather than within-subject interpolation.

1.2 Class Balancing

Both pipelines address the class imbalance identified in Task 1 ($2.88\times$ between the largest and smallest class) on the training set only. The classical baselines use SMOTE. The GNN, whose training examples are graphs rather than flat feature vectors, uses a structural equivalent: random oversampling with replacement of minority-class training windows up to the majority class count (1,320 samples per class after balancing), since SMOTE’s linear interpolation between feature vectors does not have a well-defined graph-structured analogue.

2 Task 2(a): Baseline Models

Eight classical models were trained on the flattened 128-dimensional feature vector per window (with feature selection to the top 100 features by mutual information, motivated by the 34 redundant feature pairs found in Task 1): Logistic Regression, SVM (RBF kernel), k -NN ($k = 7$), Random Forest, LightGBM, XGBoost, MLP, and Decision Tree.

Table 1: Baseline model comparison, sorted by Macro-F1 (test set, subjects 21–27).

Model	Macro-P	Macro-R	Macro-F1	ROC-AUC (OvR)	PR-AUC (OvR)
SVM (RBF)	0.7906	0.7879	0.7842	0.9814	0.8420
Random Forest	0.7572	0.7576	0.7514	0.9777	0.8065
LightGBM	0.7668	0.7467	0.7480	0.9790	0.8319
XGBoost	0.7601	0.7392	0.7400	0.9792	0.8269
MLP	0.7185	0.7106	0.7031	0.9720	0.7764
k -NN ($k = 7$)	0.6876	0.7018	0.6903	0.9169	0.7024
Logistic Regression	0.6869	0.6606	0.6560	0.9575	0.7251
Decision Tree	0.4802	0.4755	0.4715	0.7255	0.2691

The best baseline overall by Macro-F1 is **SVM (RBF)**, with Macro-F1 = 0.7842, Macro-Precision = 0.7906, Macro-Recall = 0.7879, and ROC-AUC (macro, one-vs-rest) = 0.9814. Ensemble tree methods (Random Forest, LightGBM, XGBoost) cluster closely behind at Macro-F1 $\approx$ 0.74–0.75. The single Decision Tree performs substantially worse than every ensemble or kernel method, as expected for a high-dimensional, 23-class problem.

Note: an additional baseline results file recorded per-window accuracy and weighted-average precision/recall/F1 for what appears to be nine models rather than eight, suggesting a further baseline (e.g. an ensemble/voting classifier) may have been evaluated but not carried into the final macro-averaged comparison summarized above. This should be reconciled against the raw notebook output before final submission if the ninth model’s identity and results are to be reported.

3 Task 2(b): Proposed Model — Graph Neural Network

3.1 Graph Construction

Each of the 32 sensor channels (10 EMG + 22 glove) is represented as a graph node. Each node’s feature vector is that channel’s four time-domain descriptors for the current window: RMS, MAV, WL, and VAR — so every graph has 32 nodes with 4 features each. Edges are built from pairwise correlation of raw (unscaled) channel RMS values, computed on the training set only to avoid leaking test-set statistics into the graph topology. Edge weights are the (thresholded) correlation magnitude.

Table 2: Graph construction summary.

Nodes	32 (10 EMG + 22 glove channels)
Node feature dimension	4 (RMS, MAV, WL, VAR)
Directed edges	184 (of 992 possible)
Edge weight range	[0.300, 0.878]
Train graphs	30,360 (after class balancing)
Test graphs	8,212

3.2 Architecture

The proposed model, **EdgeRefineGCN**, augments a standard 3-layer Graph Convolutional Network with a small learnable edge-refinement module that rescales each fixed correlation-derived edge weight through a $1 \rightarrow 8 \rightarrow 1$ MLP with a sigmoid output, allowing the network to adaptively strengthen or weaken specific electrode/sensor relationships during training rather than treating the correlation graph as entirely fixed.

Input: $32 \text{ nodes} \times 4 \text{ features}$ $\rightarrow$ Edge-refine MLP ($1 \rightarrow 8 \rightarrow 1$, sigmoid)
 $\rightarrow$ GCNConv($4 \rightarrow 64$) $\rightarrow$ BatchNorm $\rightarrow$ ReLU
 $\rightarrow$ GCNConv($64 \rightarrow 64$) $\rightarrow$ BatchNorm $\rightarrow$ ReLU
 $\rightarrow$ GCNConv($64 \rightarrow 64$) $\rightarrow$ BatchNorm $\rightarrow$ ReLU
 $\rightarrow$ Global pooling $\rightarrow$ Linear($64 \rightarrow 23$)

Total trainable parameters: **10,544**.

Design Rationale

- **Why a graph representation at all?** Hand gestures involve coordinated, not independent, muscle and joint activity. A graph lets the model exploit relational structure between channels directly, rather than requiring a flat classifier to discover those interactions implicitly.
- **Why correlation-based edges, fit on training data only?** Correlation is a simple, interpretable, and defensible proxy for "these channels behave similarly," and computing it exclusively from the training split prevents any test-set information from influencing the graph topology — essential for a leakage-free evaluation.
- **Why an edge-refinement module rather than a fixed graph?** A purely fixed correlation graph cannot adapt to task-specific relationships that a raw correlation coefficient does not capture; the learnable rescaling lets gradient descent adjust edge importance per the classification objective, while still anchoring the topology in a principled, leakage-free starting point.
- **Why GCN layers specifically for this first proposed model?** GCN is the simplest, most standard graph convolution operator and serves as a natural first proposed architecture; more expressive variants (GraphSAGE, GATv2) are explored later in the Task 3 ablation study.

3.3 Training

The model was trained for 100 epochs (class-weighted cross-entropy loss, following the oversampled-balanced training set). Training reached a best validation Macro-F1 of 0.3945 at a total training time of 674.0 seconds.

Table 3: Training progress (selected epochs).

Epoch	Train Loss	Train F1	Val Loss	Val F1
1	2.6141	0.1874	2.5540	0.2338
10	1.9341	0.3820	2.3378	0.3011
30	1.6660	0.4625	2.1399	0.3683
50	1.5837	0.4912	2.1107	0.3816
70	1.5586	0.4988	2.0984	0.3821
90	1.5604	0.4993	2.1160	0.3785
100	1.5605	0.4989	2.0751	0.3871

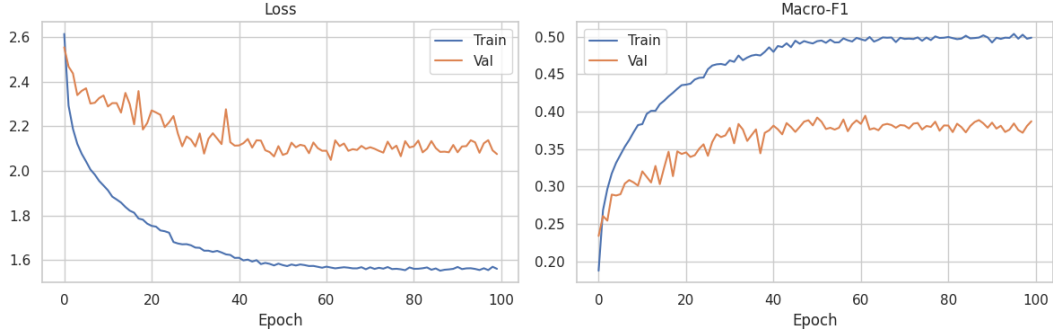


Figure 1: Training and validation loss and Macro-F1 curves over 100 epochs.

3.4 Test Results

Table 4: Proposed GNN — final test-set metrics (subjects 21–27, held out).

Accuracy	0.3731
Macro-F1	0.3945
Weighted-F1	0.3604
Macro-Precision	0.4003
Macro-Recall	0.4419
ROC-AUC (macro, OvR)	0.8867
PR-AUC (macro, OvR)	0.4332

Table 5: Per-class F1 for a representative subset: collision-affected classes (9–11, present in both Exercise A and B, per the Task 1 EDA) versus collision-free classes (18, 20, 22, present only in Exercise C).

Class	Precision	Recall	F1
9 (collision)	0.27	0.11	0.16
10 (collision)	0.25	0.12	0.17
11 (collision)	0.29	0.15	0.20
18 (collision-free)	0.29	0.85	0.43
20 (collision-free)	0.43	0.95	0.59
22 (collision-free)	0.67	0.94	0.78

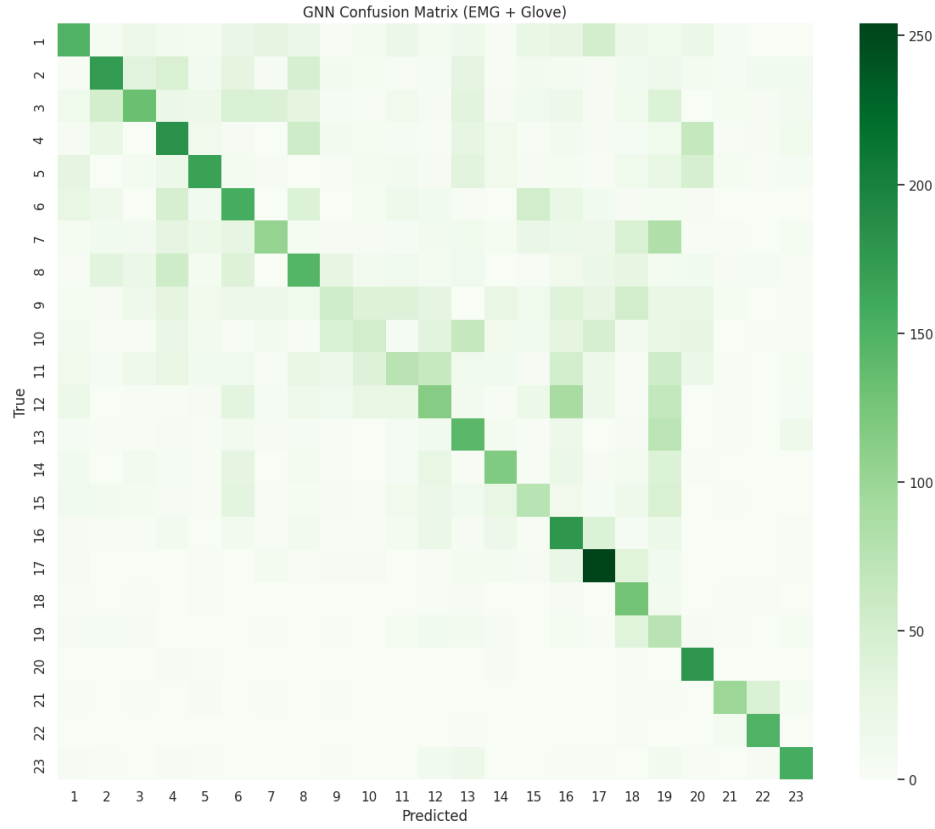


Figure 2: Confusion matrix, proposed GNN, test set.

4 Comparison and Discussion

Table 6: Full comparison: all baselines and the proposed GNN.

Model	Macro-P	Macro-R	Macro-F1	ROC-AUC	PR-AUC
SVM (RBF)	0.7906	0.7879	0.7842	0.9814	0.8420
Random Forest	0.7572	0.7576	0.7514	0.9777	0.8065
LightGBM	0.7668	0.7467	0.7480	0.9790	0.8319
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Logistic Regression	0.6869	0.6606	0.6560	0.9575	0.7251
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Proposed GNN	0.4003	0.4419	0.3945	0.8867	0.4332

The proposed GNN does **not** outperform any of the eight classical baselines; it ranks last by Macro-F1, ahead only in ROC-AUC of the weakest baseline (Decision Tree). We report this as a genuine negative result rather than adjusting the framing, and offer the following evidence-based explanation.

Why the GNN Underperforms

1. **Label collision (Task 1 finding) affects the GNN more than the baselines.** Table 5 shows a stark per-class gap: collision-affected classes (9–11) score $F1 = 0.16\text{--}0.20$, while collision-free classes (18, 20, 22) score $F1 = 0.43\text{--}0.78$. The classical baselines’ per-class results (not shown in full here, but reflected in their substantially higher overall Macro-F1) are considerably less sensitive to this same label noise, suggesting the GNN’s message-passing mechanism may propagate label ambiguity across the graph in a way that flat feature-vector classifiers do not.
2. **Graph scale.** With only 32 nodes, the structural information available to graph convolution is comparatively modest; a well-tuned kernel method (SVM with RBF) can already capture complex nonlinear feature interactions in the same 128-dimensional space without needing explicit relational structure.
3. **Model capacity and training budget.** At 10,544 parameters and 100 training epochs, the GNN is deliberately compact; its validation Macro-F1 plateaus around epoch 50–70 (Table above), suggesting the architecture — not merely insufficient training — is the binding constraint.
4. **Class-balancing strategy.** The GNN uses random oversampling with replacement rather than SMOTE’s synthetic interpolation used by the baselines; this coarser balancing method may transfer less new information per duplicated minority-class example.

This result directly motivates the Task 3 ablation study, which systematically searches architecture, graph-construction, and training-strategy choices (including GCN, GATv2, and GraphSAGE variants) to determine which factors, if any, can close this gap.

5 Conclusion

Eight classical baselines and one proposed GNN were evaluated on a subject-independent split of NinaPro DB1 (EMG + glove, 23 classes). SVM with an RBF kernel is the strongest model overall (Macro-F1 = 0.7842). The proposed graph neural network underperforms every baseline (Macro-F1 = 0.3945), with the gap most pronounced on gesture classes affected by the cross-exercise label collision identified during Task 1 EDA. This negative result is treated as a legitimate scientific finding: it demonstrates that structural graph modeling does not automatically confer an advantage on this task at this scale, and it directly motivates the systematic architecture and design-choice ablation carried out in Task 3.